

Vithursan Thiruvarooran

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Skills

- **Technologies:** Playwright, JavaScript, TypeScript, Jest, JMeter, Node.js, Python, Java, PostgreSQL, GitHub Actions
- **Concepts:** Test Automation, API Testing, Performance Testing, Regression Testing, CI/CD, Agile/Scrum

Experience

Software Development Engineer in Test, OrderGrid

Sept 2023 – April 2026

- Designed and developed a scalable **Playwright-based automation** framework for end-to-end GUI regression testing, improving execution time by 50% and reducing test flakiness by 90% through parallel execution, enhanced test reliability, and maintainable framework design
- Built a **Jest-based API automation** framework to validate backend services and critical business workflows, enabling early defect detection and increasing confidence in release quality and application stability
- Developed and executed **JMeter performance and load tests** to simulate high-traffic scenarios, identify system bottlenecks, and validate application reliability during large client onboarding and peak usage periods
- Implemented GitHub Actions CI/CD workflows to **automate regression test** execution and **developed reporting dashboards** to provide visibility into test results, quality metrics, and overall system health

Application Developer Co-op, CIBC

Jan 2020 – Aug 2020

- Earned **PEGA System Architect Certification** through self-directed learning and hands-on experience supporting enterprise applications, quickly developing proficiency in platform architecture and application workflows
- Designed and developed an **Angular-based internal web application** to streamline deployment workflows for engineering teams, improving process efficiency and reducing manual effort in daily development operations
- Diagnosed and resolved **memory leaks in a production application**, collaborating with engineers to investigate root causes, implement fixes, and improve overall system stability and application performance
- Built and optimized front-end components using Pega and CSS, improving usability and application performance

Application Developer Co-op, Faculty of Arts & Science UofT

Jan 2018 – Aug 2018

- Designed and implemented solutions to upgrade the **CHASS Data Centre**, enabling ingestion and delivery of updated **Statistics Canada CANSIM datasets**, while improving data processing workflows
- Developed **ETL workflows** using Python, Perl, PostgreSQL, and Linux scripting to process, transform, and integrate large datasets for web applications, ensuring data consistency and reliable delivery across systems
- Built scripts to parse, transform, and validate datasets, ensuring reliable system integration and data consistency
- Developed **Perl- and SQL-based workflows** to process Apache log files, extract and transform usage metrics, and load analytics data into databases for reporting and trend analysis, enabling improved visibility into user activity
- Conducted system testing and identified critical functional and design issues across multiple web applications

Projects

Scrabble App

<https://vithiru.ddns.net/scrabble>

- Built a full-stack real-time Scrabble scoring app using React, TypeScript, FastAPI, Socket.IO, MongoDB, and Redis
- Implemented WebSocket-based multiplayer synchronization for live game updates, score changes, and turn management across devices
- Implemented Redis caching and session management for active multiplayer games, improving responsiveness and supporting efficient real-time synchronization between connected clients
- Built an interactive drag-and-drop board UI with move highlighting and dynamic score feedback
- Shipped as a PWA with push notifications, background sync, and seamless update handling
- Integrated OpenCV + OCR to extract board state from photos of physical Scrabble boards

Education

University of Toronto - UTSC

June 2022

Honours Bachelor of Science

Computer Science Specialist Co-op, Software Engineering